

A Norm-Coset Reduction and a Certified Exceptional Interval in a Cyclic Cubic Field

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Abstract

Let $K = \mathbb{Q}(\theta)$, where $\theta^3 - 3\theta - 1 = 0$, and order the three real embeddings so that $\sigma_0(\theta)$ is the positive root. For $H > 0$ we study

$$M(H) = \min_{0 \neq \alpha \in \mathcal{O}_K, |\sigma_0(\alpha)| \leq H} \max(|\sigma_1(\alpha)|, |\sigma_2(\alpha)|).$$

We prove two results. First, every optimizer has absolute norm 1 or 3. Equivalently, every optimizer belongs to one of the two multiplicative layers

$$\mathcal{O}_K^\times \quad \text{or} \quad (\theta - 1)\mathcal{O}_K^\times.$$

The proof combines a rigorous log-lattice covering certificate with the arithmetic gap in possible small norms. Second, we give a globally complete exact certificate for the nonunit

$$\beta = 1 + 5\theta + 3\theta^2 = (\theta - 1)(\theta + 1)^3.$$

If L is the largest root of $x^3 - 21x^2 + 3$ and U is the largest root of $x^3 - 24x^2 + 3x + 1$, then

$$L \leq H < U \implies \operatorname{argmin} M(H) = \{\pm\beta\}.$$

At $H = U$ the unique minimizers up to sign are $\gamma = 2 + 6\theta + 3\theta^2 = (\theta + 1)^3$. The key completeness step proves that any optimizer in this transition window must lie among exactly 454 coefficient triples; the accompanying verifier checks all comparisons with rational interval arithmetic and uses no floating-point decisions.

1 The optimization problem

Let

$$f(x) = x^3 - 3x - 1, \quad K = \mathbb{Q}(\theta), \quad f(\theta) = 0.$$

The polynomial is irreducible by the rational-root test and has discriminant 81, so K is a cyclic totally real cubic field. Write its real roots as

$$r_1 < r_2 < r_0,$$

where

$$\begin{aligned} r_0 &= 1.879385241571816\dots, \\ r_1 &= -1.532088886237956\dots, \\ r_2 &= -0.347296355333861\dots, \end{aligned}$$

and set $\sigma_i(\theta) = r_i$.

For $\alpha \neq 0$, define

$$F(\alpha) = \max(|\sigma_1(\alpha)|, |\sigma_2(\alpha)|).$$

At height $H > 0$ we minimize F subject to a one-sided archimedean constraint:

$$M(H) = \min_{0 \neq \alpha \in \mathcal{O}_K, |\sigma_0(\alpha)| \leq H} F(\alpha). \quad (1)$$

The asymmetric role of σ_0 is deliberate. It makes the problem different from the usual symmetric height minimization and produces discrete transitions between unit and nonunit layers.

1.1 The full ring of integers

We will use the power basis without qualification.

Lemma 1.1. $\mathcal{O}_K = \mathbb{Z}[\theta]$.

Proof. The discriminant of the order $\mathbb{Z}[\theta]$ is 81. If $I = [\mathcal{O}_K : \mathbb{Z}[\theta]]$, then $I^2 \mid 81$ and $\text{disc}(K) = 81/I^2$. If $I > 1$, then $\text{disc}(K)$ would be 9 or 1. Minkowski's discriminant bound for a totally real cubic field gives

$$\sqrt{|\text{disc}(K)|} \geq \frac{3^3}{3!} = \frac{9}{2},$$

so $|\text{disc}(K)| > 20$. Hence $I = 1$. □

The Minkowski embedding of $\mathcal{O}_K$ is a lattice in $\mathbb{R}^3$. Once one feasible point is known, the intersection of that lattice with a bounded box is finite, so the minimum in (1) exists. A uniform feasible unit will be supplied below.

2 A global norm-coset reduction

For $\alpha \in K^\times$ put

$$p(\alpha) = \log |\sigma_0(\alpha)|, \quad q(\alpha) = \frac{\log |\sigma_1(\alpha)| - \log |\sigma_2(\alpha)|}{2}.$$

If u is a unit, then the product formula gives

$$\log F(u) = -\frac{p(u)}{2} + |q(u)|. \quad (2)$$

Thus for $B = \log H$ and a feasible unit $p(u) \leq B$,

$$\log(F(u)\sqrt{H}) = \frac{B - p(u)}{2} + |q(u)|. \quad (3)$$

This turns the search for a uniformly good unit into a covering problem in two dimensions.

Let

$$\varepsilon = \theta + 1.$$

Both θ and ε are units. Their (p, q) vectors are approximately

$$\begin{aligned} v_1 &= (0.630944724202046, 0.742104451473912), \\ v_2 &= (1.057576813574935, -0.102156317414579). \end{aligned}$$

Their determinant is $-0.849287450646\dots \neq 0$, so they generate a rank-two sublattice Λ_0 of the unit log lattice.

For $R > 0$, define the one-sided triangle

$$T_R = \{(x, y) : x \geq 0, x/2 + |y| \leq R\}.$$

Lemma 2.1 (Certified unit cover). *For every $z \in \mathbb{R}^2$ there exists $\lambda \in \Lambda_0$ such that*

$$z - \lambda \in T_{9/10}.$$

Consequently, for every $H > 0$ there is a unit $u \in \langle \theta, \varepsilon \rangle$ satisfying

$$|\sigma_0(u)| \leq H, \quad F(u) \leq \frac{e^{9/10}}{\sqrt{H}}. \quad (4)$$

Proof. Reduce z modulo Λ_0 to the centered fundamental parallelogram

$$P = \{sv_1 + tv_2 : -\frac{1}{2} \leq s, t \leq \frac{1}{2}\}.$$

It is enough to prove that for every $e \in P$, one of

$$e, \quad e + v_1, \quad e + v_2, \quad e + v_1 + v_2$$

lies in $T_{9/10}$.

The accompanying verifier proves this by exact interval arithmetic. It first isolates the three roots of f in rational intervals of width 10^{-30} . Logarithms are then enclosed by the exact series

$$\log x = 2 \sum_{k=0}^N \frac{z^{2k+1}}{2k+1} + R_N, \quad z = \frac{x-1}{x+1},$$

with a rational geometric bound on R_N . The square $[-\frac{1}{2}, \frac{1}{2}]^2$ in (s, t) coordinates is divided into 16^2 rational rectangles. On every cell, interval evaluation certifies one of the four displayed shifts. The largest cell-wise upper bound obtained is

$$0.892330436453852\dots < 0.9.$$

No floating-point value is used in a pass/fail decision. Applying the result to $z = (\log H, 0)$ and using (3) gives (4). $\square$

We next identify a gap in the possible small absolute norms.

Lemma 2.2 (Small-norm gap). *If $0 \neq \alpha \in \mathcal{O}_K$, then*

$$|N(\alpha)| \in \{1, 3\} \quad \text{or} \quad |N(\alpha)| \geq 8.$$

Proof. Modulo each of 2, 5, 7, the cubic f has no root and is therefore irreducible. By Theorem 1.1, Dedekind factorization applies directly, so each of 2, 5, 7 is inert. If one of these primes divides $N(\alpha)$, then the corresponding prime ideal divides (α) ; inertness forces $(p) \mid (\alpha)$ and hence $p^3 \mid N(\alpha)$. Therefore absolute norms 2, 4, 5, 6, 7 cannot occur. Since norms of nonzero algebraic integers are nonzero rational integers, the conclusion follows. $\square$

Theorem 2.3 (Global norm-coset reduction). *For every $H > 0$, every optimizer in (1) has absolute norm 1 or 3. More precisely,*

$$\operatorname{argmin} M(H) \subseteq \mathcal{O}_K^\times \cup (\theta - 1)\mathcal{O}_K^\times.$$

Proof. For any feasible α ,

$$|\mathbf{N}(\alpha)| = |\sigma_0(\alpha)\sigma_1(\alpha)\sigma_2(\alpha)| \leq HF(\alpha)^2,$$

so

$$F(\alpha) \geq \sqrt{\frac{|\mathbf{N}(\alpha)|}{H}}. \quad (5)$$

By Theorem 2.1, a feasible unit satisfies $F(u) \leq e^{9/10}H^{-1/2}$. On the other hand,

$$\log \sqrt{8} = \frac{3}{2} \log 2 > 1 > \frac{9}{10};$$

for example $\log 2 = 2(1/3 + 1/(3^3 \cdot 3) + \dots) > 2/3$. Hence $e^{9/10} < \sqrt{8}$, and (5) excludes every element with absolute norm at least 8 from being optimal. By Theorem 2.2, an optimizer has absolute norm 1 or 3.

It remains to identify the norm-3 layer. Set $\delta = \theta - 1$. Then $\mathbf{N}(\delta) = 3$, and from $f(\delta + 1) = 0$,

$$\delta^3 = 3(1 - \delta^2).$$

Moreover

$$1 - \delta^2 = \theta(2 - \theta), \quad (2 - \theta)(\theta + 1)^2 = 1,$$

so $1 - \delta^2$ is a unit. Thus, as ideals,

$$(3) = (\delta)^3.$$

There is a unique prime ideal above 3, namely (δ) . If $|\mathbf{N}(\alpha)| = 3$, then (α) is a prime ideal of norm 3, hence $(\alpha) = (\delta)$, and therefore $\alpha = \delta u$ for a unit u . $\square$

Theorem 2.3 is a structural statement: the global optimizer problem is exactly a competition between the unit lattice and one translated unit lattice. It does not, by itself, determine every switching height.

3 The certified exceptional interval

Define

$$\beta = 1 + 5\theta + 3\theta^2 = (\theta - 1)(\theta + 1)^3$$

and

$$\gamma = 2 + 6\theta + 3\theta^2 = (\theta + 1)^3.$$

Their norms are

$$\mathbf{N}(\beta) = -3, \quad \mathbf{N}(\gamma) = -1,$$

so β is a nonunit and γ is a unit.

Let

$$P_\beta(x) = x^3 - 21x^2 + 3, \quad P_\gamma(x) = x^3 - 24x^2 + 3x + 1.$$

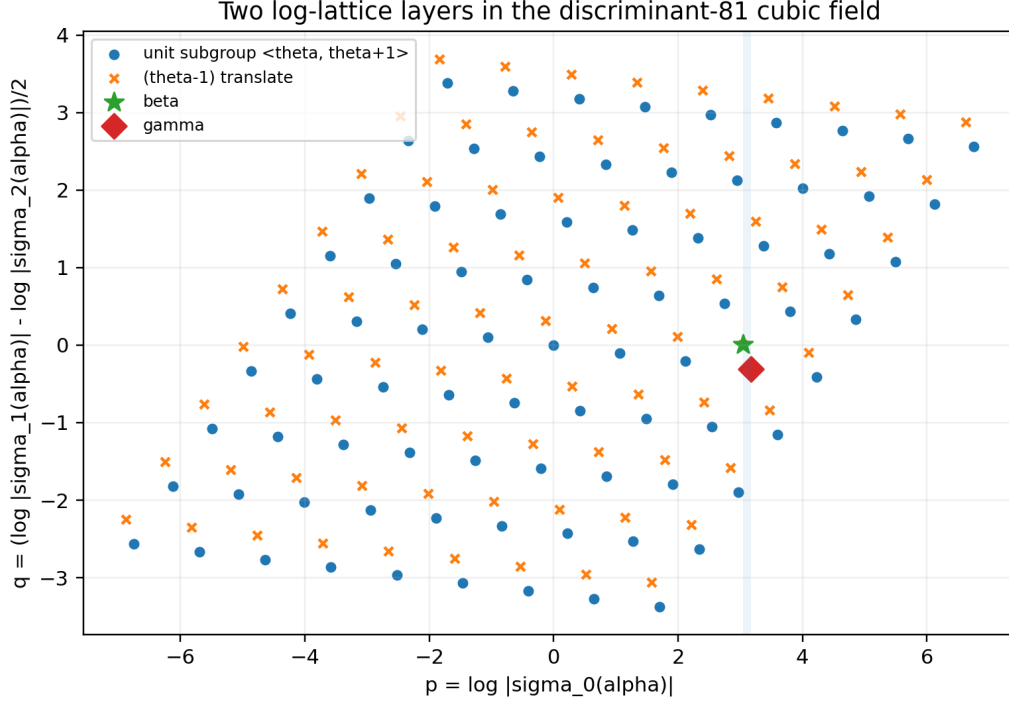


Figure 1: The subgroup generated by θ and $\theta + 1$ in (p, q) coordinates, together with the translate by $\theta - 1$. The highlighted points are $\beta = (\theta - 1)(\theta + 1)^3$ and $\gamma = (\theta + 1)^3$. The shaded vertical band marks $\log L \leq p < \log U$.

Direct reduction modulo $\theta^3 - 3\theta - 1$ verifies $P_\beta(\beta) = P_\gamma(\gamma) = 0$. Let L and U denote the largest real roots of these polynomials. Certified intervals are

$$\begin{aligned} 20.993192866572952051755448676572 < L < 20.993192866572952051755448676589, \\ 23.872578108144768819863667231221 < U < 23.872578108144768819863667231239. \end{aligned}$$

Under our ordering of embeddings,

$$L = \sigma_0(\beta), \quad U = \sigma_0(\gamma).$$

Also set

$$B = F(\beta), \quad G = F(\gamma),$$

with

$$\begin{aligned} 0.381444634811801741086373255053 < B < 0.381444634811801741086373255068, \\ 0.278066143281385509454744814816 < G < 0.278066143281385509454744814825. \end{aligned}$$

In particular $G < B$.

The finite certificate becomes global because a putative optimizer in this window has sharply bounded coefficients.

Lemma 3.1 (Completeness of the 454-triple box). *Suppose $L \leq H \leq U$, and suppose $\alpha = a + b\theta + c\theta^2 \in \mathcal{O}_K$ satisfies*

$$|\sigma_0(\alpha)| \leq U, \quad F(\alpha) \leq B.$$

Then

$$|a| \leq 2, \quad |b| \leq 6, \quad |c| \leq 3.$$

Proof. Write $y_i = \sigma_i(\alpha)$. Lagrange interpolation at the roots r_i of f gives, with $D_i = f'(r_i) = 3r_i^2 - 3$,

$$\begin{aligned} c &= \sum_{i=0}^2 \frac{y_i}{D_i}, \\ b &= \sum_{i=0}^2 \frac{r_i y_i}{D_i}, \\ a &= \sum_{i=0}^2 \frac{y_i}{r_i D_i}. \end{aligned}$$

Using $|y_0| \leq U$ and $|y_1|, |y_2| \leq B$, exact rational interval evaluation gives

$$|a| < 2.150102039486393, \quad |b| < 6.101094622358899, \quad |c| < 3.381632943042300.$$

The coefficients are integers by Theorem 1.1, so the asserted integral bounds follow. $\square$

There are exactly

$$(2 \cdot 2 + 1)(2 \cdot 6 + 1)(2 \cdot 3 + 1) - 1 = 5 \cdot 13 \cdot 7 - 1 = 454$$

nonzero triples in this box.

Theorem 3.2 (Exact global exceptional interval). *For every H with*

$$L \leq H < U,$$

the unique minimizers up to sign are $\pm\beta$:

$$\operatorname{argmin} M(H) = \{\pm\beta\}.$$

At the right endpoint,

$$\operatorname{argmin} M(U) = \{\pm\gamma\}.$$

Consequently, β is a global optimizer exactly on $[L, U)$, and every H in this interval is exceptional in the sense that the optimizer is a nonunit.

Proof. Take $L \leq H < U$. The element β is feasible, so $M(H) \leq B$. If α is any optimizer, then

$$|\sigma_0(\alpha)| \leq H < U, \quad F(\alpha) \leq B.$$

By Theorem 3.1, α lies in the 454-triple box.

The exact verifier evaluates every one of those triples on certified rational intervals for the three roots. Among all candidates with $|\sigma_0(\alpha)| < U$, the only two whose objective is at most B are β and $-\beta$; every other candidate has a rigorously separated lower objective bound greater than the rigorous upper bound for B . This proves the first assertion.

At $H = U$, γ is feasible and has objective $G < B$. The same coefficient bound remains valid. Exact enumeration shows that every one of the other 452 triples has objective strictly greater than G . Hence the unique minimizers up to sign are $\pm\gamma$.

Below L , β is infeasible. At and above U , the permanently feasible unit γ has strictly smaller objective. Therefore the full optimizer interval of β is exactly $[L, U)$. $\square$

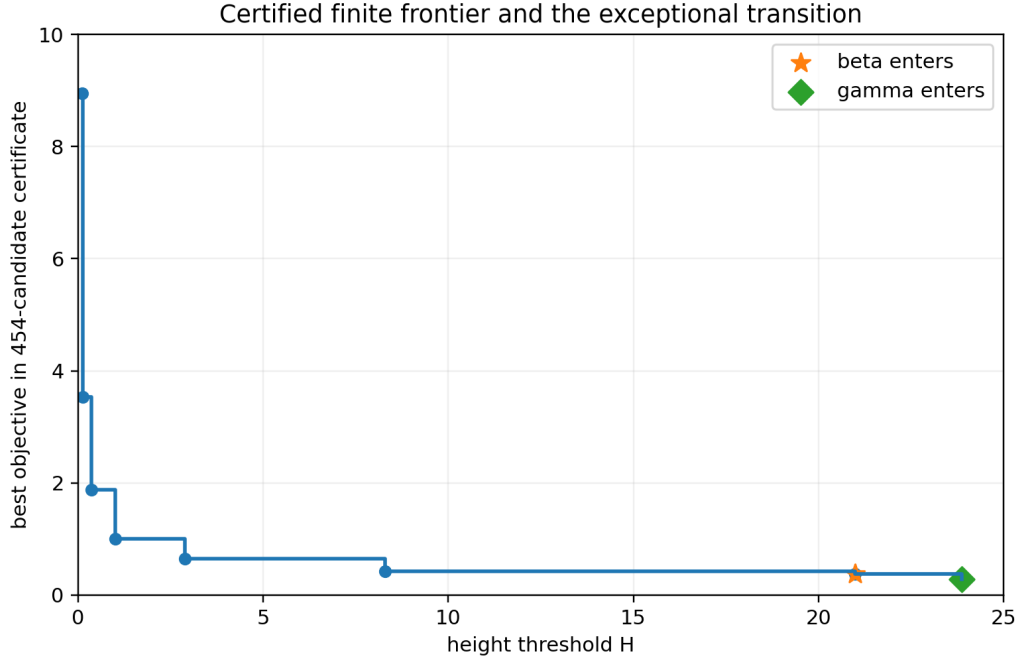


Figure 2: The record frontier inside the complete 454-triple certificate relevant to Theorem 3.2. The theorem does not infer global behavior outside the certified transition window from this finite plot.

4 What is proved, and what remains open

The results above separate three levels that were previously easy to conflate.

1. **Global structural theorem.** Every optimizer is a unit or a $(\theta - 1)$ -multiple of a unit; this is Theorem 2.3.
2. **Global exact transition.** The nonunit β is the unique optimizer up to sign on the exact interval $[L, U)$, and γ takes over at U ; this is Theorem 3.2.
3. **Distributional theory.** Numerical work motivating this project suggests a logarithmic exceptional density near 0.01945222156 and an oscillatory ordinary sampling law. Those claims are *not* proved in this release. A proof still requires a complete analysis of how the two unit-coset layers alternate over all logarithmic heights.

The norm-coset reduction materially narrows the third problem: no other norm layer can create a hidden optimizer. What remains is a two-lattice switching problem rather than an unrestricted search over all algebraic integers.

5 Verification protocol

The release includes a standalone script, `verify.py`. Its pass/fail decisions use exact integers and rational numbers only. The verifier checks:

- isolation of the three roots of $x^3 - 3x - 1$;

- the identities defining β , γ , and the ramified norm-3 layer;
- the 16-by-16 log-lattice covering certificate, with logarithms bounded by rational series remainders;
- the inverse-Vandermonde coefficient bounds proving completeness of the 454-triple box;
- all 454 finite comparisons in Theorem 3.2; and
- the inertness tests modulo 2, 5, and 7 used in the norm gap.

The discovery calculations and the verifier are deliberately separated. A reader can rerun the verifier without trusting the numerical search that originally suggested β .

6 Relation to classical cubic-field geometry

The field in this paper is the $a = 0$ member of Shanks' simplest cubic family. The general family, its units, and its cyclic structure were studied by Shanks [1]. Voronoi reduction, relative minima, and the geometry of cubic ideals give a broad classical framework for understanding special lattice points in cubic fields; a modern treatment is given by Hambleton and Williams [2]. Fundamental-parallelepiped methods also appear in recent work on indecomposable integers in totally real cubic fields, for example [4, 3].

The one-sided constrained minimax functional (1) is not the standard relative-minimum problem. The present manuscript therefore makes only the claims proved above and does not assert that the general geometric viewpoint is new. A full novelty assessment would require a more exhaustive comparison with the literature on relative minima, reduction, and inhomogeneous approximation in cubic fields.

7 Conclusion

The main obstruction in this project was not finding an unusual candidate; it was proving that the candidate could not be defeated by an algebraic integer outside the search window. The inverse-Vandermonde argument removes that obstruction for the principal transition, upgrading a finite search into the exact global interval theorem Theorem 3.2. Independently, the log-lattice covering argument and the small-norm arithmetic reduce the entire optimization problem to two multiplicative layers, Theorem 2.3. These results provide a rigorous base from which the remaining density and switching questions can be attacked without hidden norm layers or unverified finite-search assumptions.

References

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